RS59 Yield, Efficiency & Cross-Section Comparison

Existing vs. Latest Results

October 28, 2026

Single Differential Cross-Section Formalism ($\mathrm{d}\sigma/\mathrm{d}p_T$)

LH_2 (pp) Single Differential Cross Section:

$$\frac{\mathrm{d}\sigma_{pp}}{\mathrm{d}p_T} = \frac{1}{\epsilon_{\mathrm{acc}}\mathcal{L}\Delta p_T}\left[\frac{Y^{\mathrm{LH2}}_{\mathrm{total}} - Y^{\mathrm{LH2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LH2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LH2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right] \tag{1}$$

LD_2 (pd) Single Differential Cross Section:

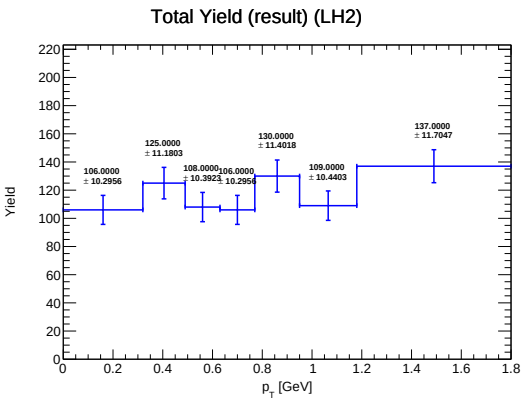
$$\begin{aligned}\frac{\mathrm{d}\sigma_{pd}}{\mathrm{d}p_T} = \frac{1}{\epsilon_{\mathrm{acc}}\mathcal{L}\Delta p_T}\Bigg(&\left[\frac{Y^{\mathrm{LD2}}_{\mathrm{total}} - Y^{\mathrm{LD2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LD2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LD2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right] \\ &- \frac{T_{HD}}{T_{HH}}\frac{I_{\mathrm{LD2}}}{I_{\mathrm{LH2}}}\left[\frac{Y^{\mathrm{LH2}}_{\mathrm{total}} - Y^{\mathrm{LH2}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{LH2}}_{\mathrm{signal}}\rangle} - \frac{I_{\mathrm{LH2}}}{I_{\mathrm{flask}}}\left(\frac{Y^{\mathrm{flask}}_{\mathrm{total}} - Y^{\mathrm{flask}}_{\mathrm{mixed}}}{\langle\epsilon^{\mathrm{flask}}_{\mathrm{signal}}\rangle}\right)\right]\Bigg)\end{aligned} \tag{2}$$

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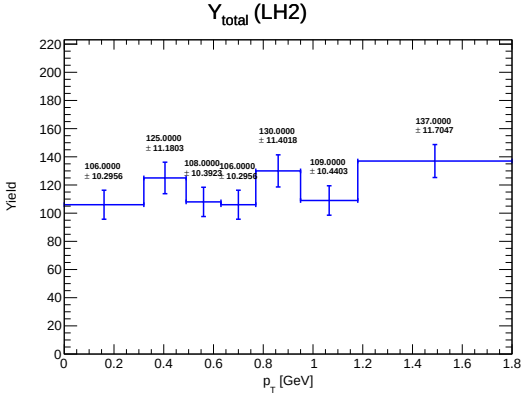
1. Yield Distributions
2. Efficiency Distributions
3. Corrected & Subtracted Yields
4. Cross Sections & Ratios
5. Systematic Uncertainties

LH_2 Target: Total Yield (Y_total_LH2.pdf)

Existing Results

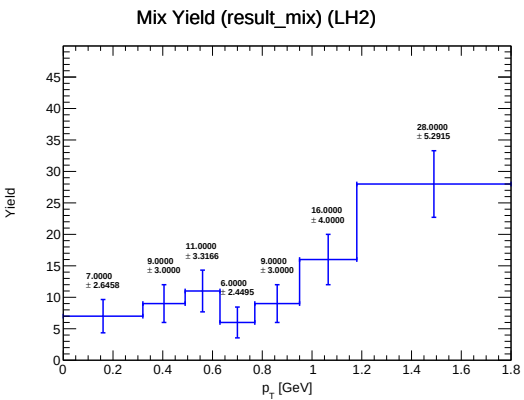


Latest Results

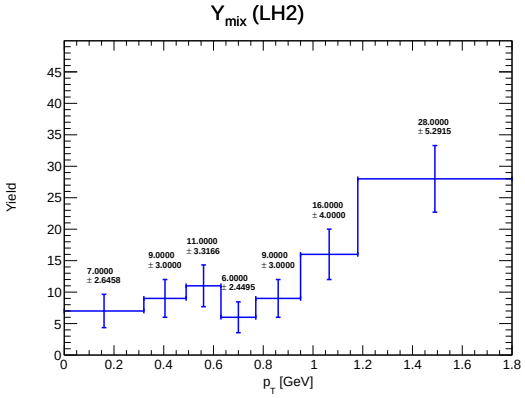


LH_2 Target: Mixed Yield (Y_mix_LH2.pdf)

Existing Results

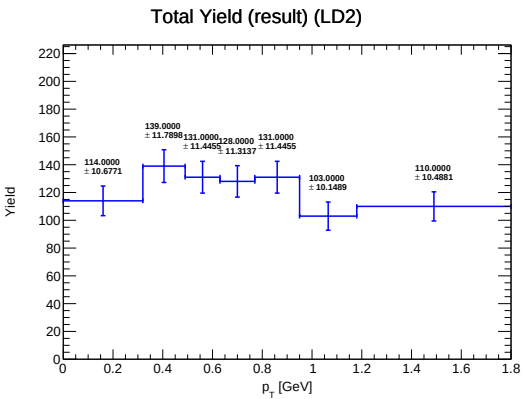


Latest Results

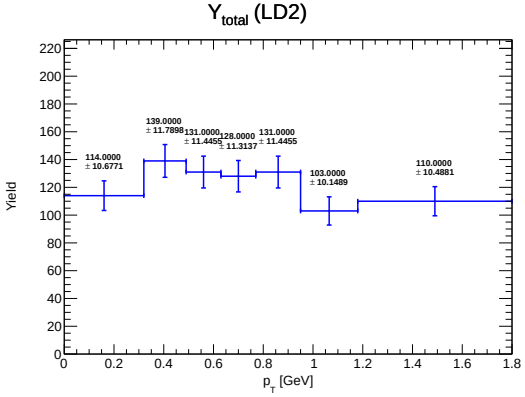


LD_2 Target: Total Yield (Y_total_LD2.pdf)

Existing Results

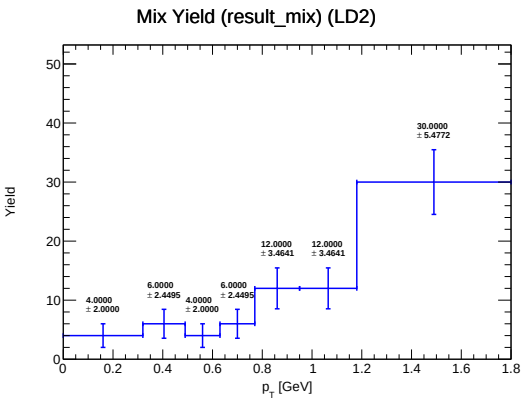


Latest Results

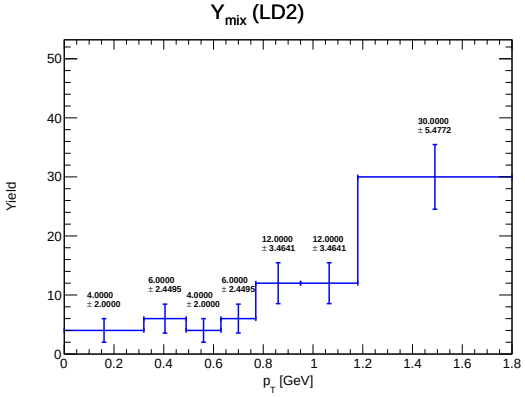


LD_2 Target: Mixed Yield (`Y_mix_LD2.pdf`)

Existing Results

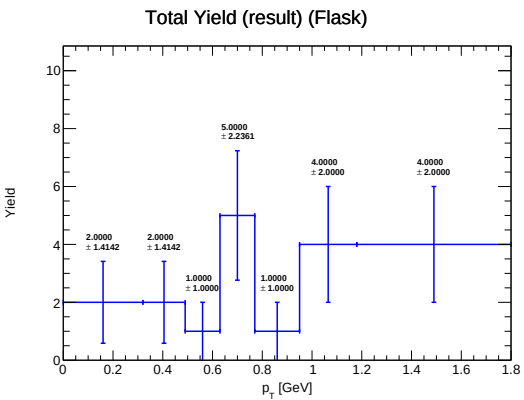


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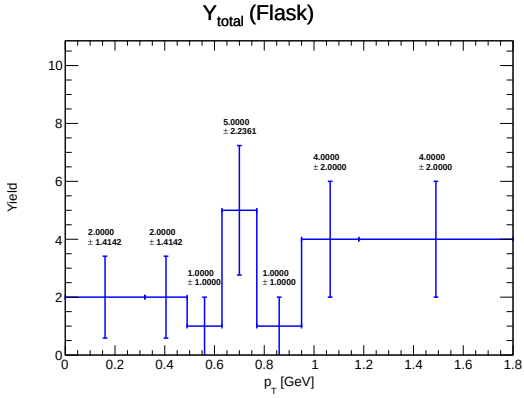


Empty Flask: Total Yield (Y_total_Flask.pdf)

Existing Results

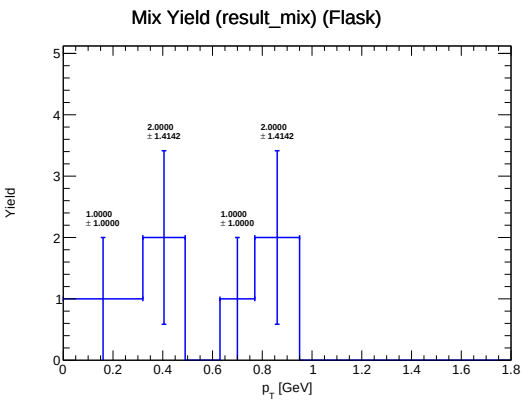


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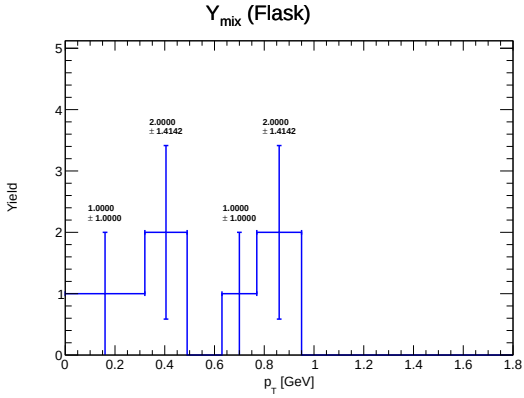


Empty Flask: Mixed Yield (Y_mix_Flask.pdf)

Existing Results

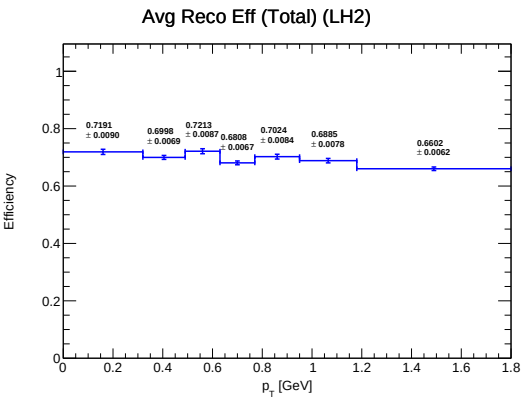


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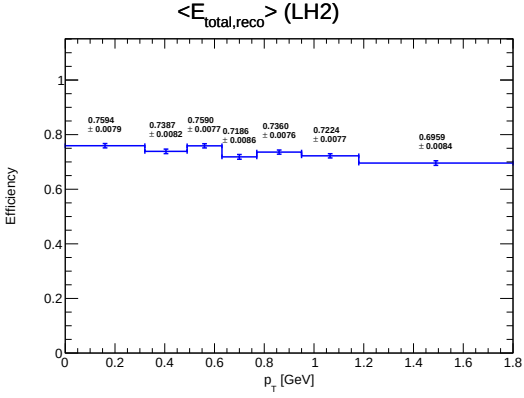


LH_2 Target: Total Reco Efficiency (E_total_reco_LH2.pdf)

Existing Results

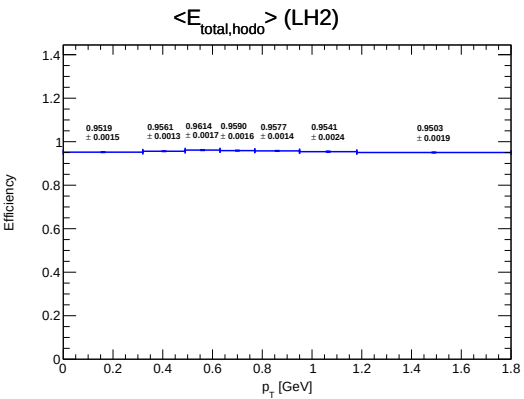


Latest Results

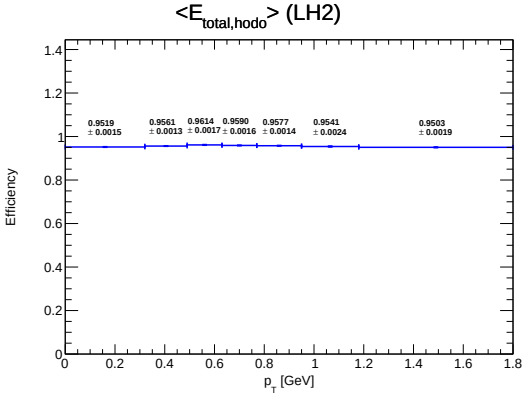


LH_2 Target: Total Hodo Efficiency (E_total_hodo_LH2.pdf)

Existing Results

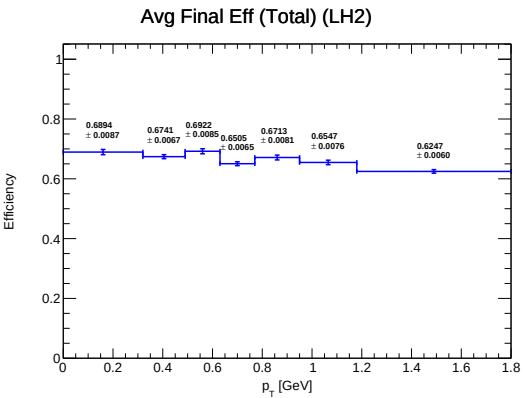


Latest Results

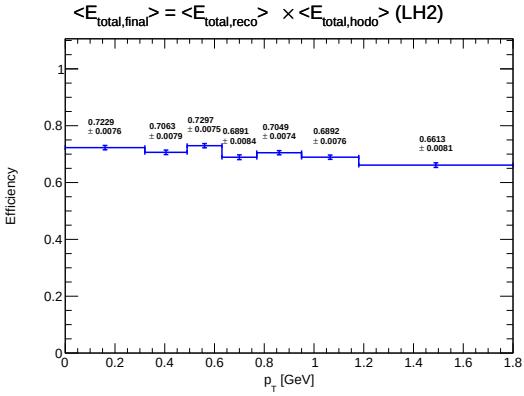


LH_2 Target: Total Final Efficiency (E_total_final_LH2.pdf)

Existing Results

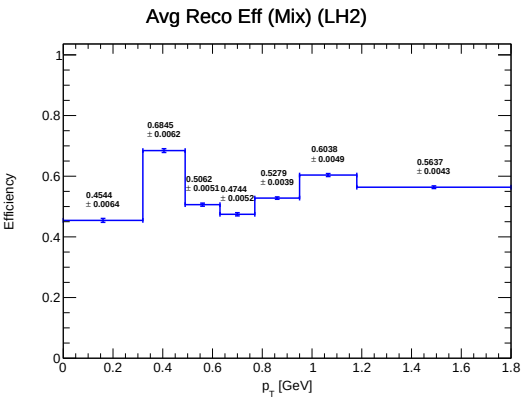


Latest Results

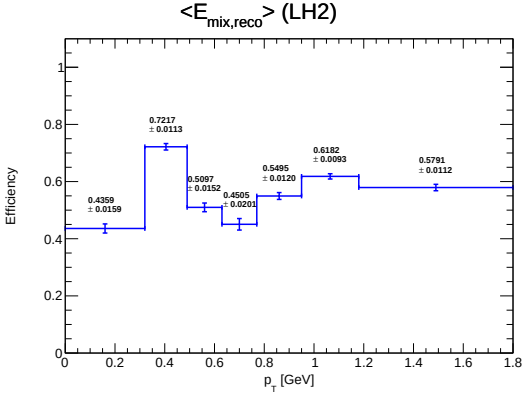


LH_2 Target: Mixed Reco Efficiency (E_mix_reco_LH2.pdf)

Existing Results

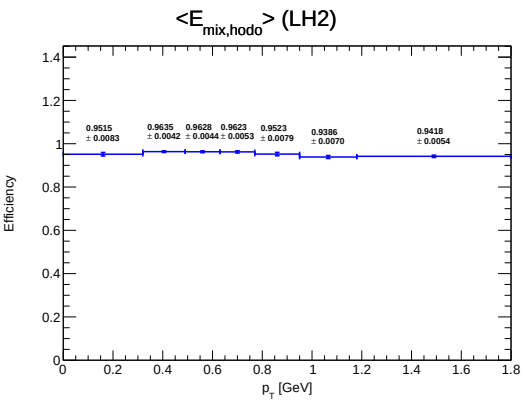


Latest Results

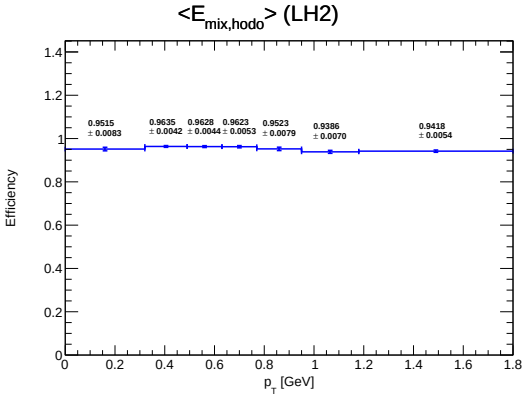


LH_2 Target: Mixed Hodo Efficiency (E_mix_hodo_LH2.pdf)

Existing Results

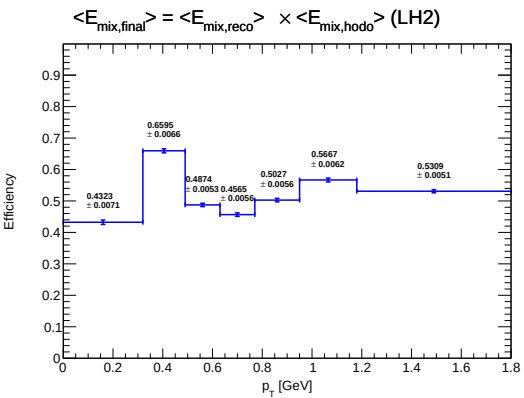


Latest Results

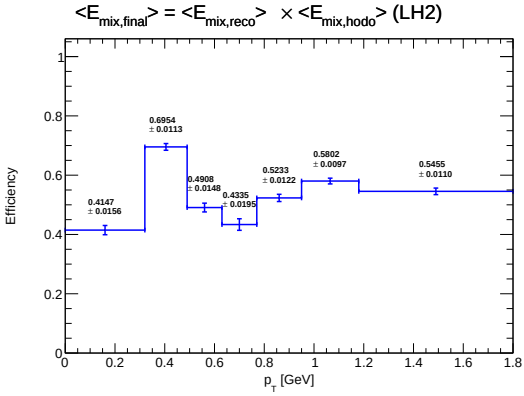


LH_2 Target: Mixed Final Efficiency (E_mix_final_LH2.pdf)

Existing Results

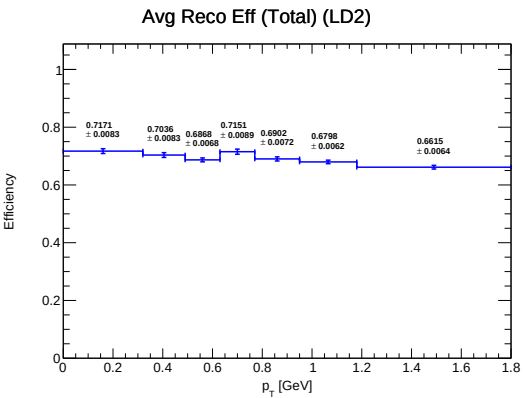


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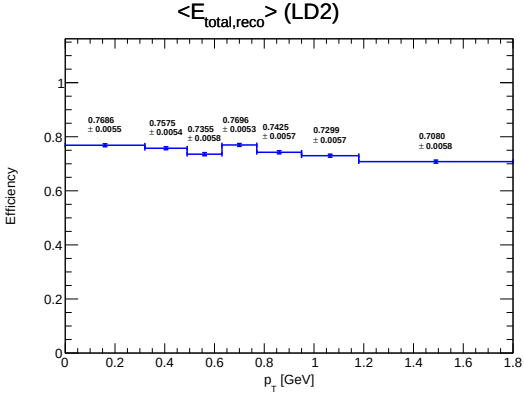


LD_2 Target: Total Reco Efficiency (E_total_reco_LD2.pdf)

Existing Results

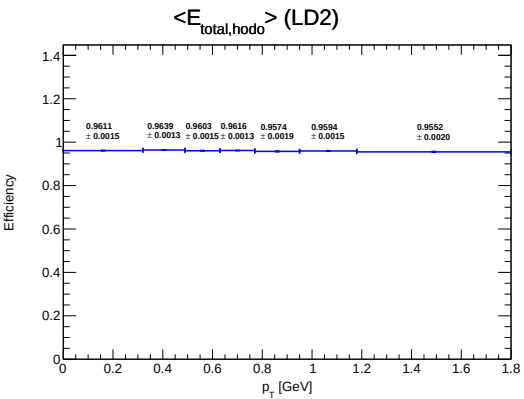


Latest Results

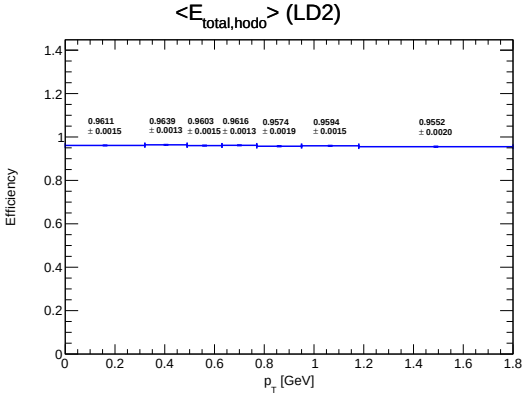


LD_2 Target: Total Hodo Efficiency (E_total_hodo_LD2.pdf)

Existing Results

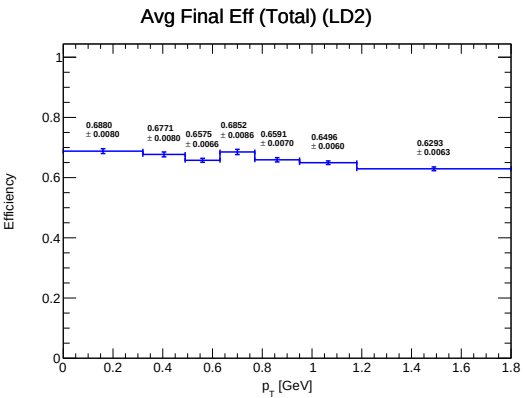


Latest Results

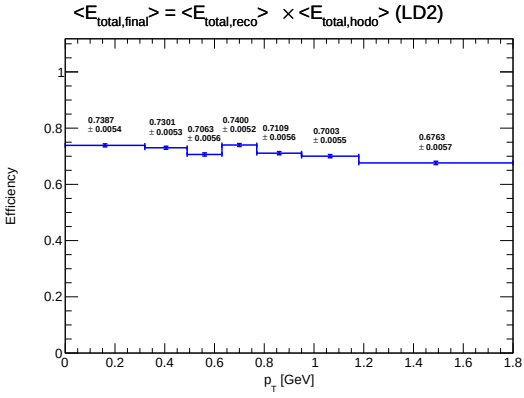


LD_2 Target: Total Final Efficiency (E_total_final_LD2.pdf)

Existing Results

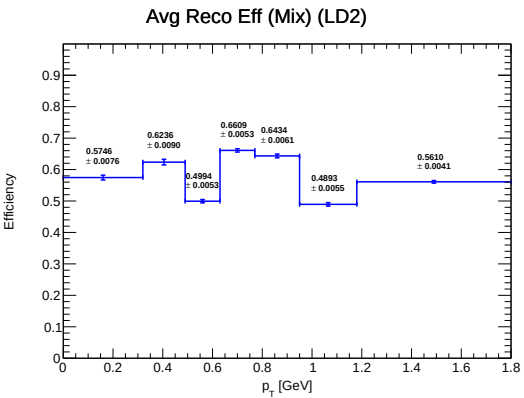


Latest Results

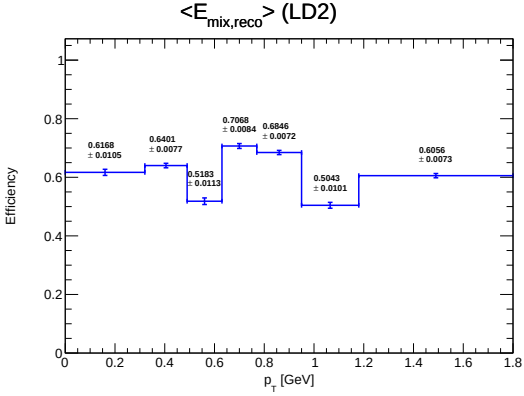


LD_2 Target: Mixed Reco Efficiency (E_mix_reco_LD2.pdf)

Existing Results

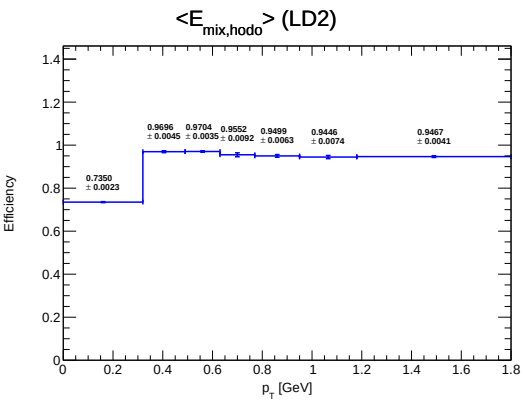


Latest Results

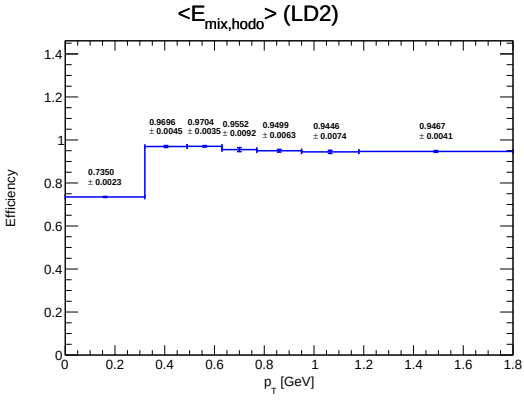


LD_2 Target: Mixed Hodo Efficiency (E_mix_hodo_LD2.pdf)

Existing Results

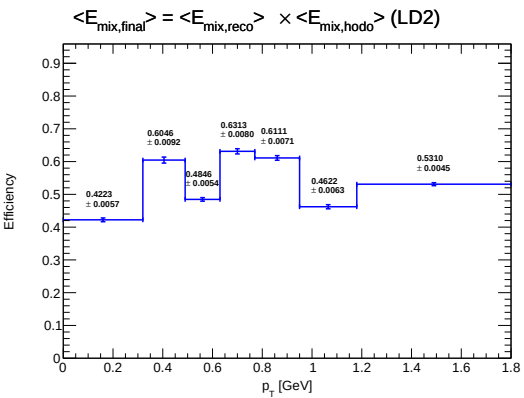


Latest Results

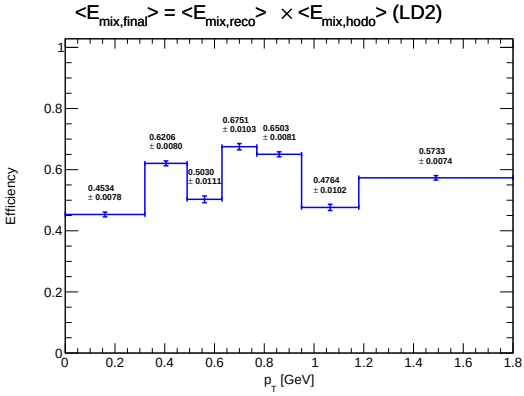


LD_2 Target: Mixed Final Efficiency (`E_mix_final_LD2.pdf`)

Existing Results

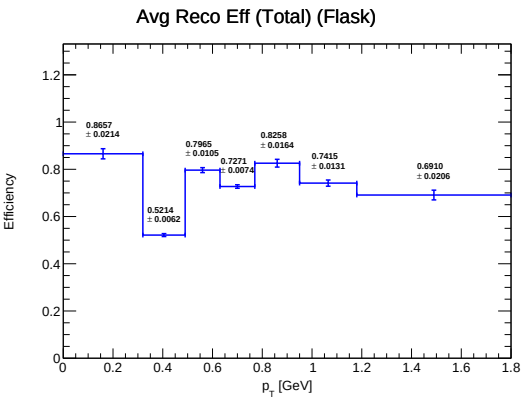


Latest Results

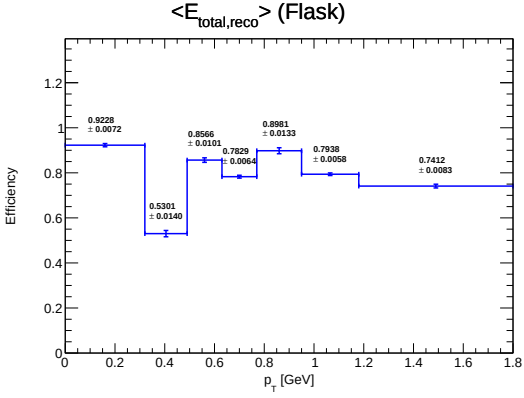


Empty Flask: Total Reco Efficiency (E_total_reco_Flask.pdf)

Existing Results

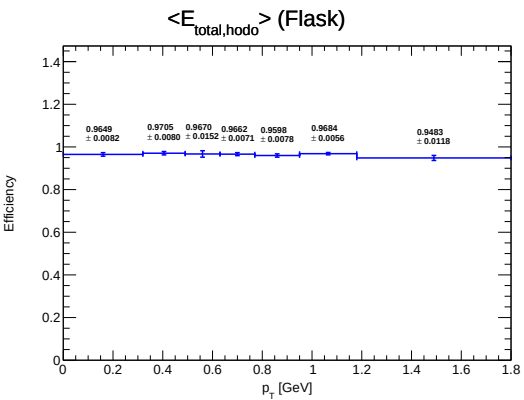


Latest Results

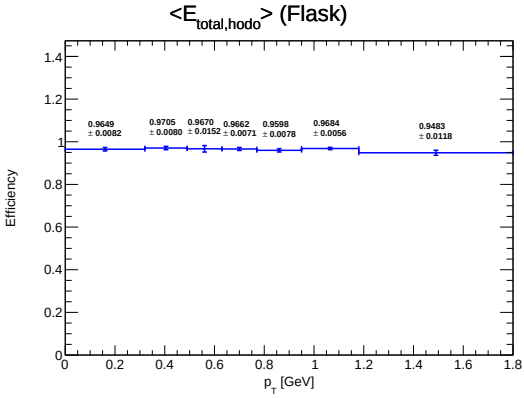


Empty Flask: Total Hodo Efficiency (E_total_hodo_Flask.pdf)

Existing Results

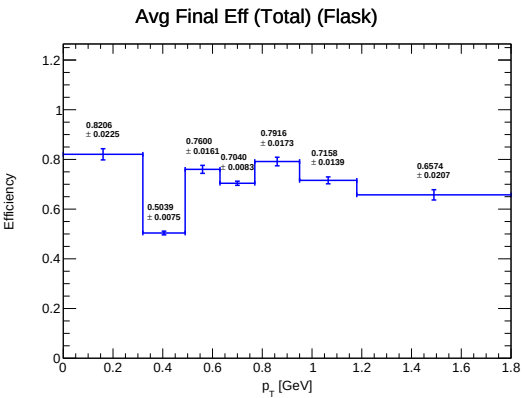


Latest Results

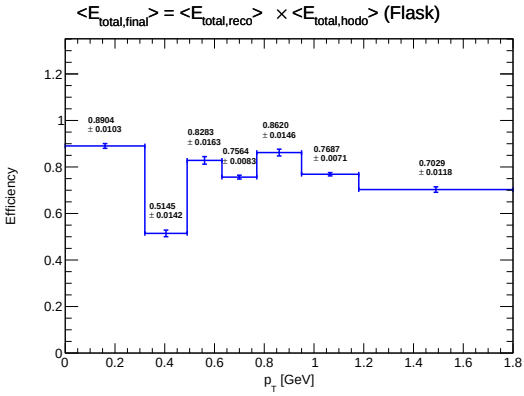


Empty Flask: Total Final Efficiency (E_total_final_Flask.pdf)

Existing Results

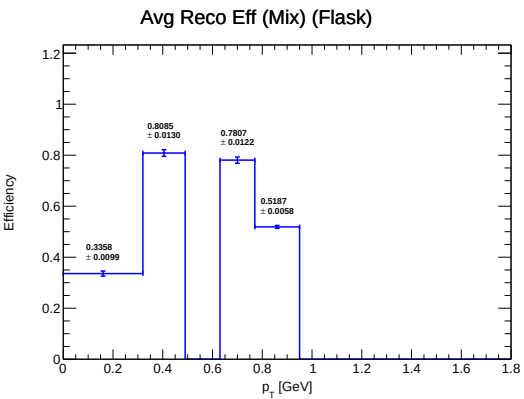


Latest Results

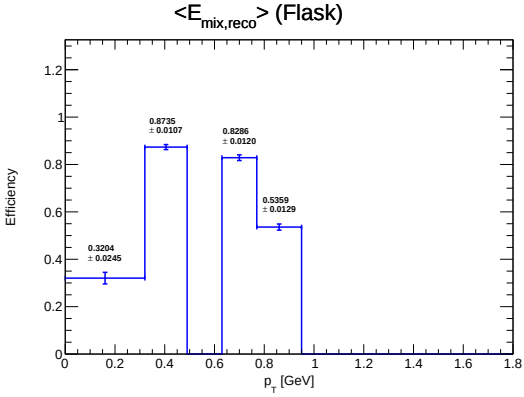


Empty Flask: Mixed Reco Efficiency (E_mix_reco_Flask.pdf)

Existing Results

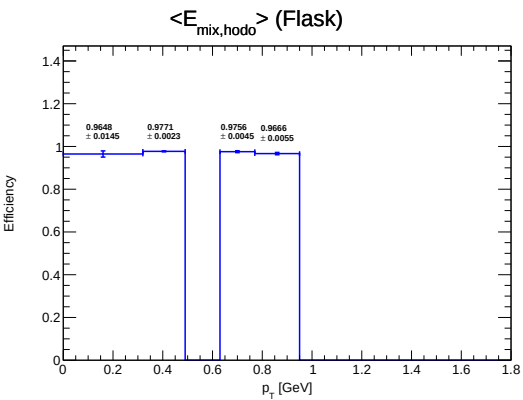


Latest Results

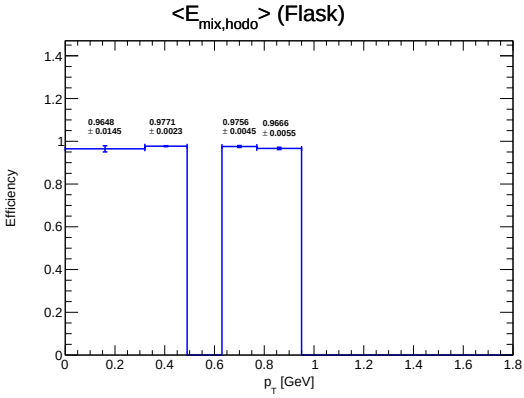


Empty Flask: Mixed Hodo Efficiency (E_mix_hodo_Flask.pdf)

Existing Results

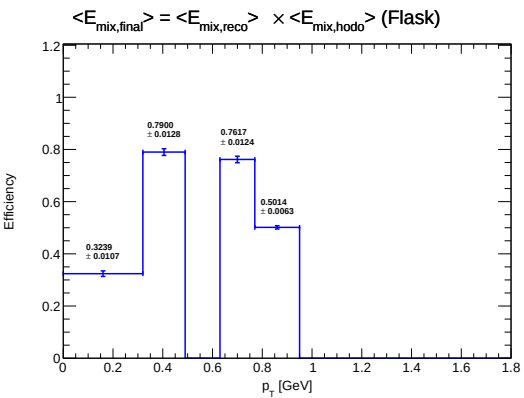


Latest Results

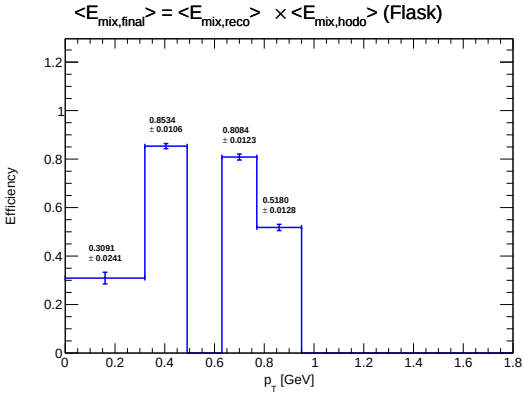


Empty Flask: Mixed Final Efficiency (E_mix_final_Flask.pdf)

Existing Results

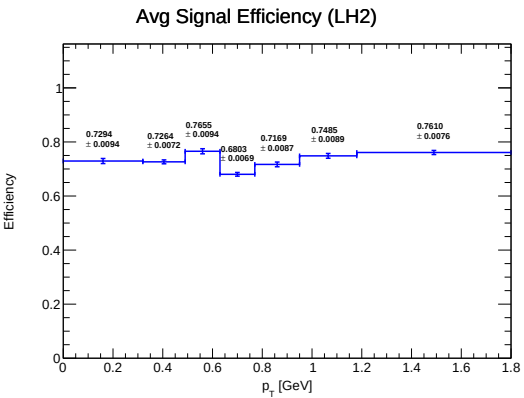


Latest Results

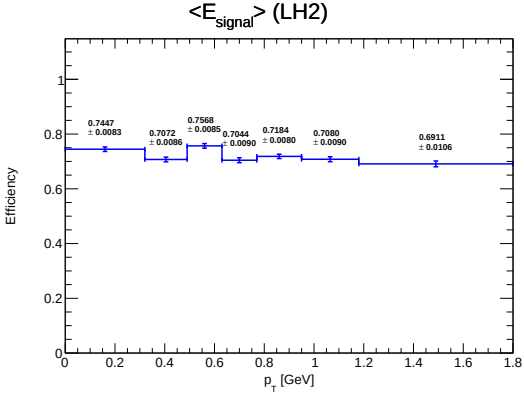


LH_2 Target: Final Signal Efficiency (E_final_signal_LH2.pdf)

Existing Results

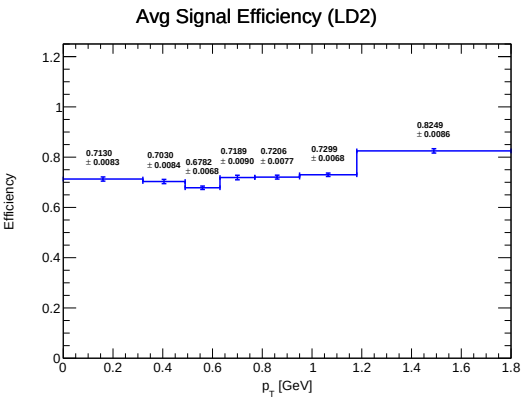


Latest Results

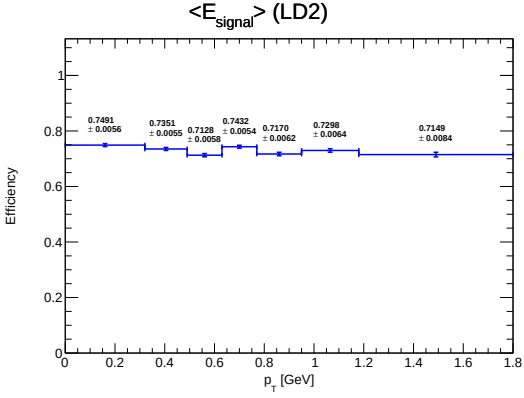


LD_2 Target: Final Signal Efficiency (E_final_signal_LD2.pdf)

Existing Results

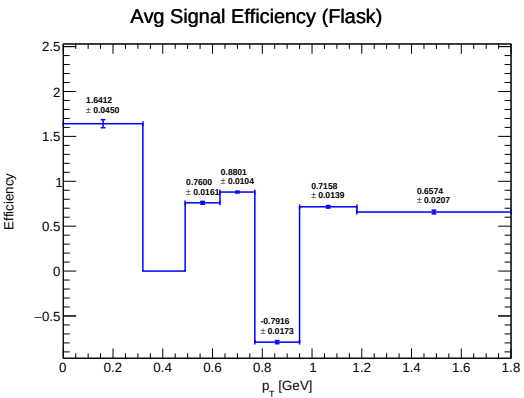


Latest Results

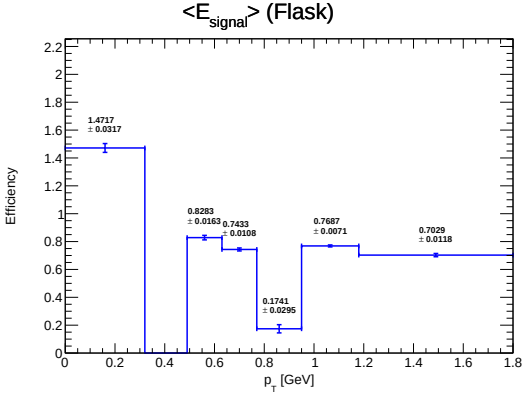


Empty Flask: Final Signal Efficiency (E_final_signal_Flask.pdf)

Existing Results

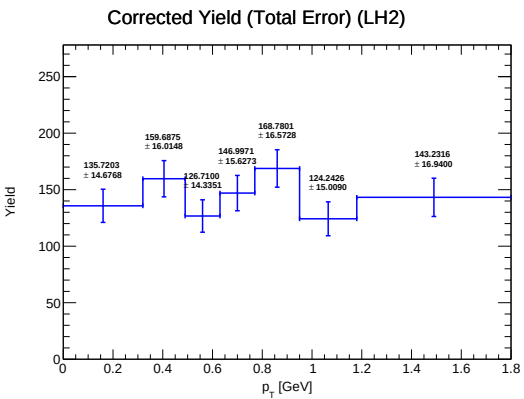


Latest Results

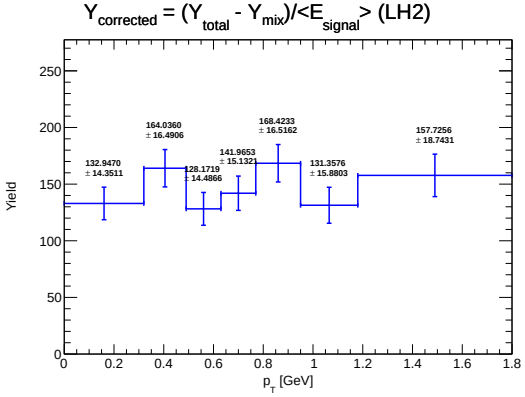


LH_2 Target: Corrected Yield (`Y_corrected_LH2.pdf`)

Existing Results

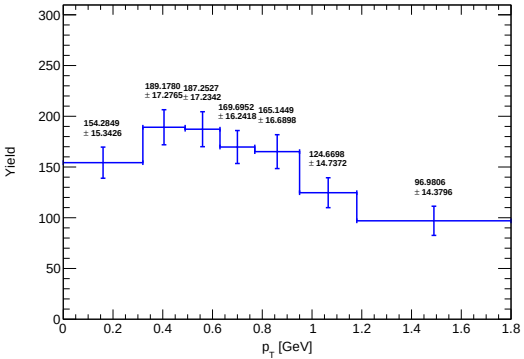


Latest Results

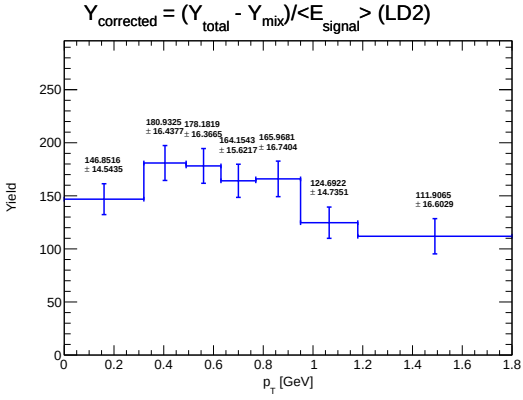


LD_2 Target: Corrected Yield (`Y_corrected_LD2.pdf`)

Existing Results

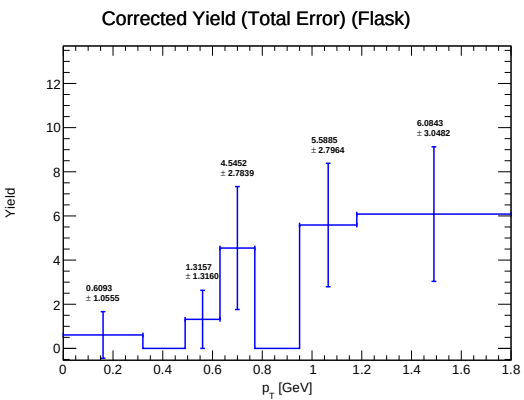


Latest Results

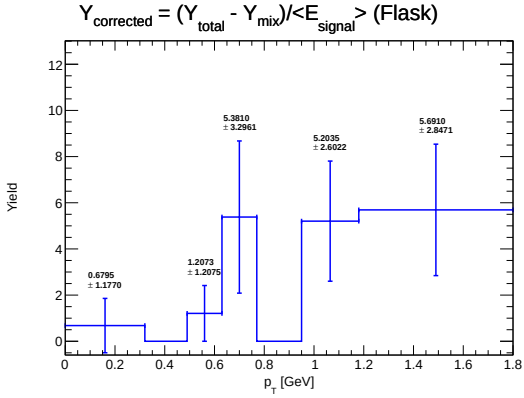


Empty Flask: Corrected Yield (Y_corrected_Flask.pdf)

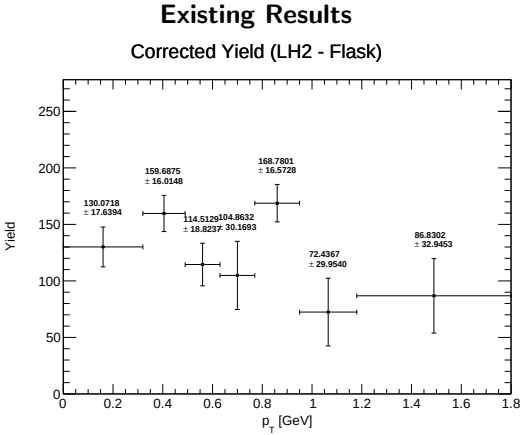
Existing Results

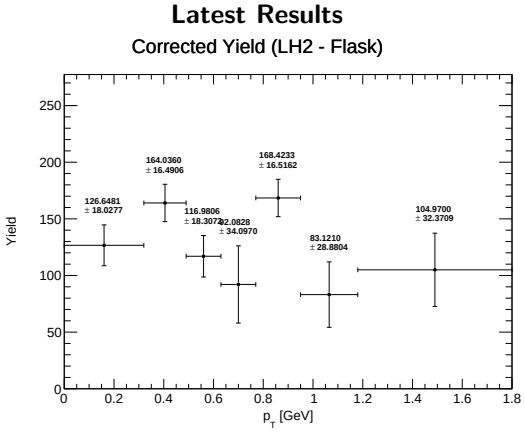


Latest Results

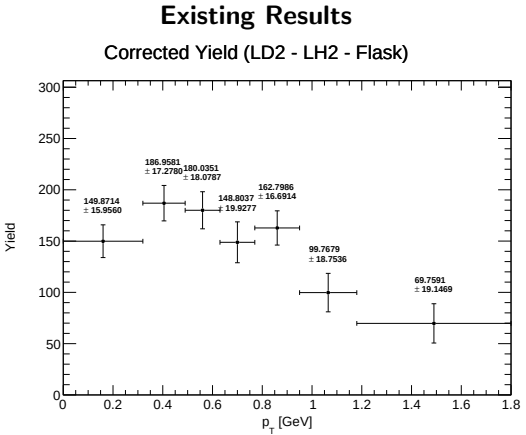


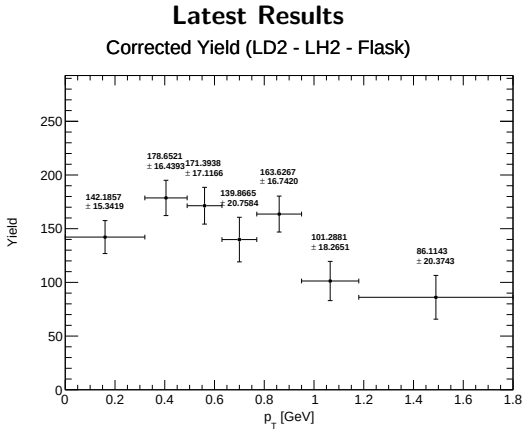
LH_2 Target: Subtracted Corrected Yield (Y_corrected_Subtracted_LH2.pdf)





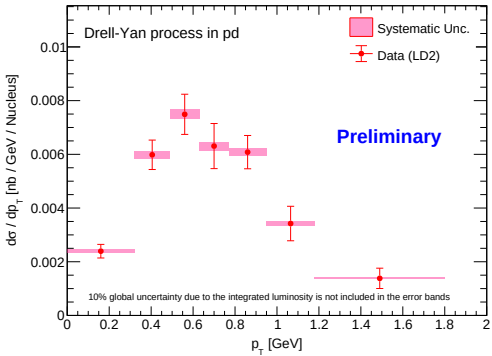
LD_2 Target: Subtracted Corrected Yield (Y_corrected_Subtracted_LD2.pdf)



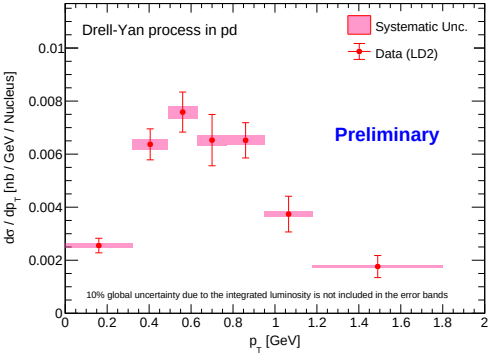


LD_2 Cross Section vs. p_T (CrossSection_LD2_geom_vs_pT.pdf)

Existing Results

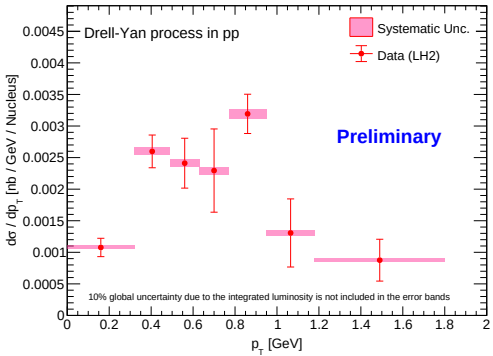


Latest Results

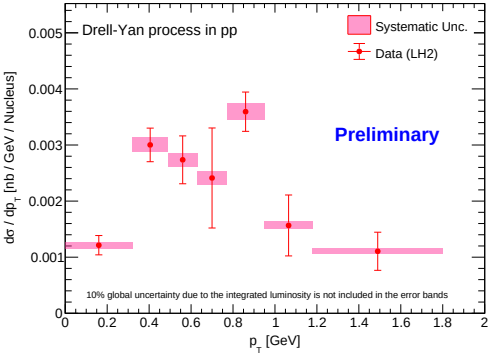


LH_2 Cross Section vs. p_T (CrossSection_LH2_geom_vs_pT.pdf)

Existing Results

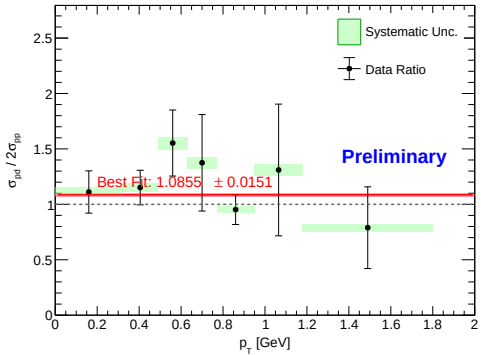


Latest Results

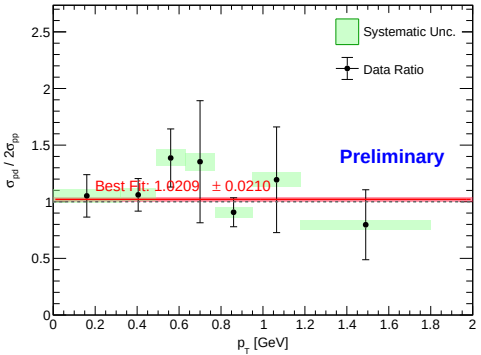


Cross Section Ratio $\sigma_{pd}/2\sigma_{pp}$ vs. p_T (CrossSection_Ratio_pd_2pp_vs_pT_geom.pdf)

Existing Results

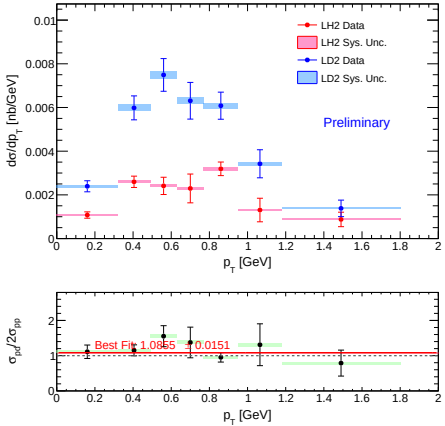


Latest Results

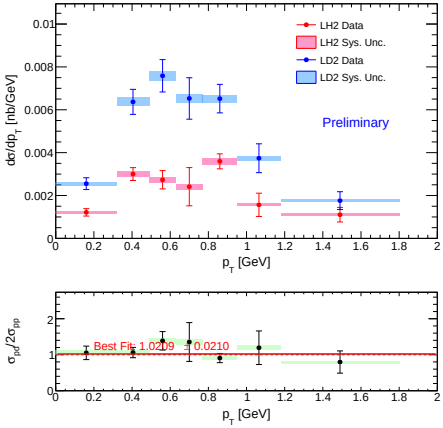


Combined Cross Section & Ratio vs. p_T (Combined_XSec_Ratio_vs_pT_geom.pdf)

Existing Results



Latest Results



LH_2 Target: Systematic Components (Table_Systematics_LH2.tex)

Latest Results

$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.016768	0.000765	0.045622	0.001829	0.000453	0.247808	126.6481	1.4798	0.011685
[0.32, 0.49)	0.016532	0.000693	0.041889	0.002900	0.000720	0.248166	164.0360	1.9950	0.012162
[0.49, 0.63)	0.015708	0.000672	0.042763	0.003874	0.000957	0.247144	116.9806	1.4615	0.012494
[0.63, 0.77)	0.014024	0.000611	0.043584	0.003034	0.001294	0.426602	92.0828	1.9440	0.021111
[0.77, 0.95)	0.013390	0.000525	0.039182	0.003310	0.000768	0.232102	168.4233	1.8648	0.011072
[0.95, 1.18)	0.011870	0.000457	0.038458	0.004258	0.000933	0.219190	83.1210	1.7350	0.020874
[1.18, 1.80)	0.007876	0.000279	0.035459	0.003692	0.000794	0.215045	104.9700	2.5686	0.024470

Existing Results

$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.019414	0.000441	0.022716	0.001829	0.000453	0.247808	130.0718	1.7494	0.013450
[0.32, 0.49)	0.018594	0.000392	0.021087	0.002900	0.000720	0.248166	159.6875	1.5921	0.009970
[0.49, 0.63)	0.017444	0.000378	0.021680	0.003874	0.000957	0.247144	114.5129	1.5817	0.013812
[0.63, 0.77)	0.016789	0.000357	0.021237	0.003034	0.001294	0.426602	104.8632	1.5648	0.014922
[0.77, 0.95)	0.015105	0.000297	0.019683	0.003310	0.000768	0.232102	168.7801	2.0514	0.012154
[0.95, 1.18)	0.012398	0.000249	0.020057	0.004258	0.000933	0.219190	72.4367	1.7850	0.024643
[1.18, 1.80)	0.008227	0.000153	0.018561	0.003692	0.000794	0.215045	86.8302	2.2809	0.026269

LD_2 Target: Systematic Components (Table_Systematics_LD2.tex)

Latest Results

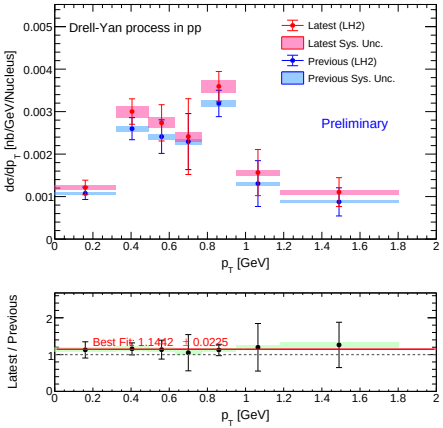
$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.018048	0.000557	0.030842	0.001829	0.000453	0.247808	142.1857	1.0999	0.007736
[0.32, 0.49)	0.017115	0.000492	0.028761	0.002900	0.000720	0.248166	178.6521	1.3609	0.007617
[0.49, 0.63)	0.016740	0.000486	0.029008	0.003874	0.000957	0.247144	171.3938	1.4614	0.008527
[0.63, 0.77)	0.015873	0.000454	0.028603	0.003034	0.001294	0.426602	139.8665	1.2425	0.008883
[0.77, 0.95)	0.014461	0.000381	0.026328	0.003310	0.000768	0.232102	163.6267	1.4434	0.008821
[0.95, 1.18)	0.012216	0.000323	0.026452	0.004258	0.000933	0.219190	101.2881	1.1155	0.011013
[1.18, 1.80)	0.008176	0.000199	0.024301	0.003692	0.000794	0.215045	86.1143	1.3678	0.015884

Existing Results

$[p_{T,\min}, p_{T,\max})$	Acc. Corr.	Acc. Err.	Acc. Frac. Err.	ψ' Corr.	ψ' Err.	ψ' Frac. Err.	Yield Corr.	Yield Err.	Yield Frac. Err.
[0.00, 0.32)	0.020298	0.000441	0.021727	0.001829	0.000453	0.247808	149.8714	1.8063	0.012052
[0.32, 0.49)	0.019066	0.000392	0.020565	0.002900	0.000720	0.248166	186.9581	2.2618	0.012098
[0.49, 0.63)	0.017807	0.000378	0.021237	0.003874	0.000957	0.247144	180.0351	1.8838	0.010463
[0.63, 0.77)	0.017477	0.000357	0.020401	0.003034	0.001294	0.426602	148.8037	2.1432	0.014403
[0.77, 0.95)	0.015424	0.000297	0.019276	0.003310	0.000768	0.232102	162.7986	1.7736	0.010894
[0.95, 1.18)	0.013143	0.000249	0.018919	0.004258	0.000933	0.219190	99.7679	1.2426	0.012455
[1.18, 1.80)	0.008446	0.000153	0.018081	0.003692	0.000794	0.215045	69.7591	1.2902	0.018495

Cross-Section Comparison vs. p_T: LH_2 and LD_2

LH_2 Plot



LD_2 Plot

